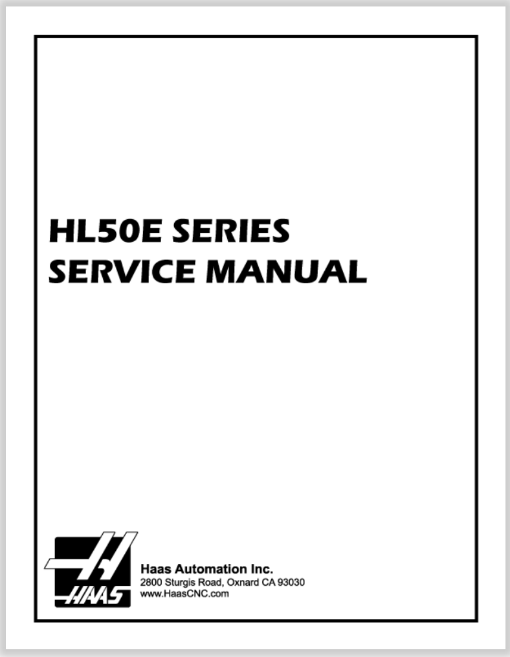
1.1 SERVICE DISCLAIMER



HL50E SERIES
SERVICE MANUAL

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Disclaimer

⚠ **Only a certified technician is allowed to repair this machine.**

⚠ **All information provided for reference and illustrative purposes only. Haas Automation cannot be held responsible for repairs you perform. Only services and repairs provided by authorized Haas Factory Outlet technician are guaranteed.**

⚠ **Only an authorized Haas Factory Outlet distributor should service or repair a Haas machine that is protected by the original factory warranty. Servicing by any other party voids the warranty.**

1.2 INTRODUCTION

This service manual will show you how to install, operate and maintain the HL-50E and the HL-50E-XT Laser Marking Machine.

1.3 SPECIFICATIONS

HL-50E Series Components

The HL-50E/XT Fiber Laser Marking Machine consists of three main components: an energy source, a laser medium, and an optical resonator.

This section provides detailed specifications for the HL50E.

HL-50E Specifications

LASER	S.A.E.	METRIC
Module	Han's Laser	Han's Laser
Power	50.0 W	50.0 W
Power Stability (8h)	<±1% rms	<±1% rms
Power Requirement	110 VAC	220 VAC
Output Power	10% to 100%	10% to 100%
Output Frequency	20 kHz to 100 kHz	20 kHz to 100 kHz
Beam Quality	M^2< 1.3	M^2 < 1.3
Wavelength	1064nm (Infrared)	1064nm (Infrared)
Repeatability	±0.0001 in	±0.003 mm
Focus Spot Diameter	<0.0004 in	<0.01 mm
Minimum Line Width	0.0004 in	0.01 mm
Marking Range	4.3 in x 4.3 in (F160 lens)	110 mm x 110 mm (F160 lens)
Marking Depth	0.047 in	1.2 mm
Marking Speed	800 std. characters/sec	800 std. characters/sec
Cooling	Air Cooled	Air Cooled

FUME EXTRACTOR	S.A.E.	METRIC
Input Power	310.0 W	310.0 W
Power Requirement	110 VAC	220 VAC
Filter Efficiency	11.8 µin, 99.97%	0.3 µm, 99.97%
System Flow Rate	21895 ft^3/hr	620 m^3/hr
Noise	<60db	<60db
Weight	43 lbs	19.5 kgs

DIMENSIONS	S.A.E.	METRIC
Overall Length (L)	43.7 in	1110 mm
Overall Width (W)	28.9 in	735 mm
Overall Height (H)	64.6 in	1640 mm
Enclosure Length (A)	39.4 in	1000 mm
Enclosure Width (B)	27.4 in	695 mm
Enclosure Height (C)	33.5 in	850 mm
Fixture Plate	17.7 in x 13.8 in	50 mm x 350 mm
Fume Extractor	21.7 in x 16.3 in x 36.2 in	550 mm x 415 mm x 920 mm

GENERAL	S.A.E.	METRIC
Country of Origin	China	China
Gross Weight	426 lbs	193 kg
Warranty	12-month	12-month

HL-50E-XT Specifications

LASER	S.A.E.	METRIC
Module	Han's Laser	Han's Laser
Power	50.0 W	50.0 W
Power Stability (8h)	<±1% rms	<±1% rms
Power Requirement	110 VAC	220 VAC
Output Power	10% to 100%	10% to 100%
Output Frequency	20 kHz to 100 kHz	20 kHz to 100 kHz
Beam Quality	M^2 < 1.3	M^2 < 1.3
Wavelength	1064nm (Infrared)	1064nm (Infrared)
Repeatability	±0.0001 in	±0.003 mm
Focus Spot Diameter	<0.0004 in	<0.01 mm
Minimum Line Width	0.0004 in	0.01 mm
Marking Range	4.3 in x 4.3 in (F160 lens)	110 mm x 110 mm (F160 lens)
Marking Depth	0.047 in	1.2 mm
Marking Speed	800 std. characters/sec	800 std. characters/sec
Cooling	Air Cooled	Air Cooled

FUME EXTRACTOR	S.A.E.	METRIC
Input Power	310.0 W	310.0 W
Power Requirement	110 VAC	220 VAC
Filter Efficiency	11.8 µin, 99.97%	0.3 µm, 99.97%
System Flow Rate	21895 ft^3/hr	620 m^3/hr
Noise	<60db	<60db
Weight	43 lbs	19.5 kgs

DIMENSIONS	S.A.E.	METRIC
Overall Length (L)	56 in	1420 mm
Overall Width (W)	57 in	1450 mm
Overall Height (H)	90.6 in	2300 mm
Enclosure Length (A)	51.2 in	1300 mm
Enclosure Width (B)	52.5 in	1333 mm
Enclosure Height (C)	78.6 in	2000 mm
Fixture Plate	39.4 in x 25.6 in	1000 mm x 650 mm
Fume Extractor	21.7 in x 16.3 in x 36.2 in	550 mm x 415 mm x 920 mm

GENERAL	S.A.E.	METRIC
Country of Origin	China	China
Gross Weight	983 lbs	446 kg
Warranty	12-month	12-month